MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If the common difference of an AP is 5, then what is $a_{18} - a_{13}$? [1]

(A) 5 (B) 20 (C) 25 (D) 30

Q2. What is the common difference of an AP in which $a_{18} - a_{14} = 32$? [1]

(A) 8 (B) -8 (C) -4 (D) 4

Q3. Two APs have the same common difference. The first term of one is -1 and that of the other is -8. Then the difference between their 4th terms is: [1]

(A) -1 (B) -8 (C) 7 (D) -9

Q4. If 7 times the 7th term of an AP is equal to 11 times its 11th term, then its 18th term will be: [1]

(A) 7 (B) 11 (C) 18 (D) 0

Q5. **Assertion (A):** If the sum of first n terms of an AP is $S_n = 3n^2 + 5n$, then its common difference is 6.
Reason (R): In an AP, the coefficient of n^2 in S_n is equal to $\frac{d}{2}$, which implies $d = 2 \times 3 = 6$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

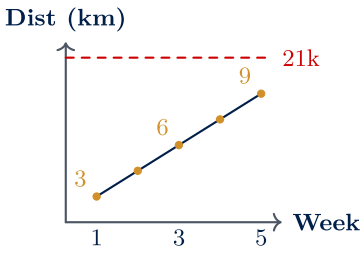
Q6. Find the 11th term from the end of the AP $10, 7, 4, \ldots, -62$. [2]

Q7. If the sum of first m terms of an AP is equal to the sum of first n terms ($m \neq n$), prove that the sum of its first $(m+n)$ terms is zero. [3]

Q8. Solve the equation: $1 + 4 + 7 + 10 + \cdots + x = 287$. [3]

Q9. If the mth term of an AP is $\frac{1}{n}$ and the nth term is $\frac{1}{m}$, show that the sum of its first mn terms is $\frac{1}{2}(mn+1)$. [3]

Q10. CASE STUDY: Athletics Half-Marathon Training Progression



Neha, a disciplined student athlete of Class 10, sets a goal to participate in the National Youth Half-Marathon event (21 km). Under her coach's advice, she designs a structured weekly progressive training schedule to systematically build cardiovascular stamina without risking physical injury.

In the first week of her training regimen, Neha runs 3 km every day. At the start of each subsequent week, she increases her daily running distance by a uniform increment of 1.5 km.

Based on the above mathematical scenario, answer the following questions:

(i) Write down the daily running distances completed by Neha during each of the first three weeks. [1]

(ii) In which week of training will her daily running target reach exactly 15 km? [1]

(iii) In which week will her scheduled daily run first reach or exceed the full half-marathon distance of 21 km? [2]

— OR —

If Neha runs 6 days per week throughout her training, calculate the total cumulative distance she runs over the first 10 weeks. [2]

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